

Meridian Casualty: Hiring Is Broken — Can You Fix It?

[PLACEHOLDER — narrative to be written. Every paragraph below is filler that shows the build pipeline works. No number in this file is real. Numbers go in only when copied from outputs/RESULTS.md with their exhibit id.]

0. Recommendation

[PLACEHOLDER — narrative to be written.] We recommend option **(b)**, **make applicants bear it**: re-time the situational section, disable paste, and flag very fast answers for recruiters. The cost against the status quo is stated as a range per retained employee and per year (E19), excluding and including operational quality (E17). Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

[PLACEHOLDER — one-sentence reason.] The section has stopped screening on the new system (E12, E12b, E12d); the timed system keeps a usable share of the signal (E12b, E12d); the timed centers retain more hires today (E01b, E17). Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

[PLACEHOLDER — the biggest hole, named.] The seven timed centers are small and their own signal is eroding (E07b, E12e), so copying the legacy configuration is an upper bound; the true gain lies between the difference-in-differences estimate and the legacy-observed gap (E19). Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo.

1. (a) What the process gets right and where it fails

[PLACEHOLDER — narrative to be written.] Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. License, referral, resume and the structured interview still predict retention in every cell and are stronger in the AI era (E11, E11b, E14). The volume machine works: openings fill and applications arrive (E10). Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem.

[PLACEHOLDER — judgment 1.] The screen still weights the essay exactly as it did when the essay worked (E11c). The process has not adapted. Ut enim ad minima veniam, quis nostrum exercitationem ullam corporis suscipit laboriosam, nisi ut aliquid ex ea commodi consequatur? Quis autem vel eum iure reprehenderit qui in ea voluptate velit esse quam nihil molestiae consequatur, vel illum qui dolorem eum fugiat quo voluptas nulla pariatur?

[PLACEHOLDER — judgment 2.] Recruiter capacity is spread thinner every year (E10, E10b, E10c); the new ATS draws more applications per opening in the AI era (E10d), so the least informative essays arrive where recruiters have the least time to read them. At vero eos et accusamus et iusto odio dignissimos ducimus qui blanditiis praesentium voluptatum deleniti atque corrupti quos dolores et quas molestias excepturi sint occaecati cupiditate non provident, similique sunt in culpa qui officia deserunt mollitia animi, id est laborum et dolorum fuga.

[PLACEHOLDER — consequence.] Retention falls more on the new ATS after 2023 (E01, E01b), reopen rates rise (E02, E02b), and separations rise in every reason category (E03, E03c). Whitfield's "best-run

centers are worst" is the pilot-wave effect (E04, E05). Et harum quidem rerum facilis est et expedita distinctio. Nam libero tempore, cum soluta nobis est eligendi optio cumque nihil impedit quo minus id quod maxime placeat facere possimus, omnis voluptas assumenda est, omnis dolor repellendus.

2. (b) Does the situational section still work as a screen, and how we know

[PLACEHOLDER — the comparison and on whom.] *Comparison*: the slope of essay score on the six-month outcome within each of four cells (legacy or new ATS, application before or after 2023-01). *On whom*: permanent hires, who were selected on the essay and so understate the slope, and flex placements, who were not selected on the essay by Meridian. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet ut et voluptates repudiandae sint et molestiae non recusandae.

[PLACEHOLDER — result.] E12 (bins), E12b (hires retention), E12c (hires reopen), E12d (flex reopen). Signal retained by cell as a range across the two samples: E15e. Itaque earum rerum hic tenetur a sapiente delectus, ut aut reiciendis voluptatibus maiores alias consequatur aut perferendis doloribus asperiores repellat. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

[PLACEHOLDER — mechanism.] Slow essays still work everywhere; fast high scores on the new ATS in the AI era predict worse retention; effort no longer earns points (E13, E13b, E13c). Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur.

[PLACEHOLDER — placebo.] Every other input holds or strengthens (E14). Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium.

[PLACEHOLDER — timing and identification.] Scores climb from 2023 in all new-ATS centers regardless of migration date (E06, E06b); no pre-trend (E15); within-center before and after (E12e); not the labor market (E16); not range restriction (E15c); survives the wild-cluster bootstrap (E15d). Totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit.

[PLACEHOLDER — what must be true.] (i) Flex placement is not selected on the essay; (ii) nothing unobserved changed at migrating centers in 2023 that also changed essay validity (E15); (iii) the 2023-01 cutoff is substantively right, and any cutoff in 2023 gives the same story (E06b). Sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet.

[PLACEHOLDER — could not determine.] Whether any individual used AI; what the unreviewed applicants look like; the legacy system's own erosion path beyond 2025. Quis autem vel eum iure reprehenderit qui in ea voluptate velit esse quam nihil molestiae consequatur.

3. (c) The decision

[PLACEHOLDER — worksheet and options.] The worksheet reproduces the prompt's worked example (E17a). Status quo and counterfactuals (E17). The four options in one table (E19), with (c) and (d) explicitly assumption-driven. At vero eos et accusamus et iusto odio dignissimos ducimus qui blanditiis praesentium voluptatum deleniti atque corrupti quos dolores et quas molestias excepturi sint occaecati cupiditate non provident.

Option	Cost per retained employee	Annual cost vs status quo	Basis
(a) Do nothing	[E19]	[E19]	observed
(b) Applicants bear it	[E19 range]	[E19 range]	observed in legacy centers
(c) Meridian bears it	[E19]	[E19]	assumed
(d) Bear it after hiring	[E19]	[E19]	assumed

[PLACEHOLDER — sensitivity and break-evens.] E20 and E20b give the sensitivity; E20c gives the extra vacancy days that erase (b)'s saving at each retention bound and the recovery share (c) would need. Similique sunt in culpa qui officia deserunt mollitia animi, id est laborum et dolorum fuga. Et harum quidem rerum facilis est et expedita distinctio.

[PLACEHOLDER — what would change our recommendation.]

- We would recommend (c) instead if [PLACEHOLDER].
- We would recommend (d) instead if [PLACEHOLDER].
- We would recommend doing nothing if [PLACEHOLDER].

[PLACEHOLDER — Round 2 triggers.] Vendor cost and feasibility of re-timing; legacy centers matched on size and region; a labor-market shock; a revised cost per reopened claim; evidence on applicant drop-off under a timed section. Nam libero tempore, cum soluta nobis est eligendi optio cumque nihil impedit quo minus id quod maxime placeat facere possimus.

Appendix: exhibits

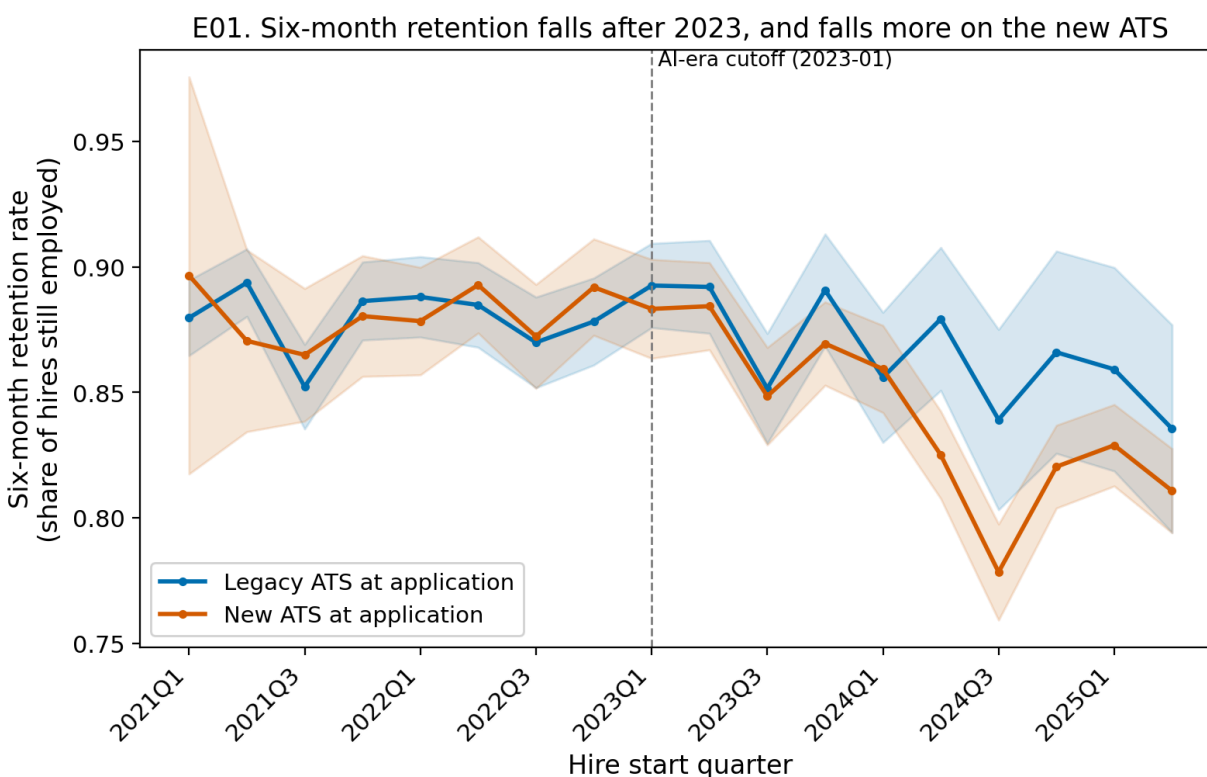
Exhibits appear in the order listed in the narrative outline. Each figure carries its own title, sample statement and source note; each table is reproduced from outputs/tables/ as written by the analysis scripts.

E00. Sample construction and data-integrity checks

check	value	expected
applications.csv rows	529302	529302
hires.csv rows	46266	46266
flex_placements.csv rows	28672	28672
center_month.csv rows	2400	2400
ats_migration.csv rows	40	40
hires_joined rows (== hires.csv)	46266	46266
flex_joined rows (== flex_placements.csv)	28672	28672
applications with outcome=hired	46266	46266
applications with outcome=flex_placement	28672	28672
hires censored (no 6mo outcome)	4589	expect ~4589
hires with observed 6mo outcome (analysis sample)	41677	expect ~41677
coaching_hours null total	4161	n/a
coaching_hours null AND censored	571	n/a
coaching_hours null AND NOT censored (true missing)	3590	n/a
hires: ATS at start != ATS at application (use application-month ATS)	530	expect ~530
application-to-start lag distribution (months)	{0: np.int64(11169), 1: np.int64(31475), 2: np.int64(3609), 3: np.int64(13)}	n/a
hires_obs by regime (cutoff 2023-01)	{'legacy': 19366, 'new_pre': 5920, 'new_post': 16391}	n/a
flex by regime (cutoff 2023-01)	{'legacy': 14620, 'new_pre': 2346, 'new_post': 11706}	n/a
centers never migrated to new ATS	7	expect 7

Sample: All five raw files, 2021-01 to 2025-12, 40 centers. *Notes:* Analysis sample for hire outcomes = hires_obs (retained_6mo observed); censored hires (started 2025-07 or later) are excluded from every hire regression and never imputed. *Source:* analysis/00_load.py

E01



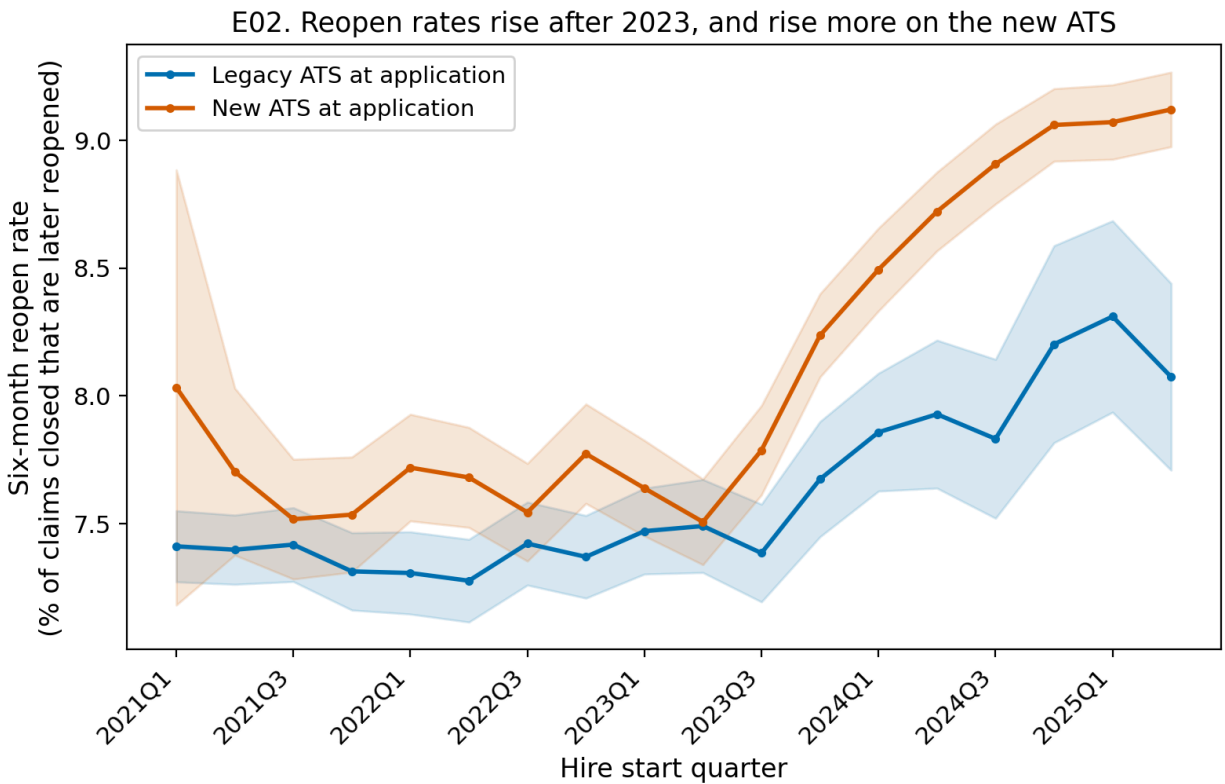
Sample: 41,677 permanent hires with an observed six-month outcome, start months 2021-01 to 2025-06 (later starts are censored).
Source: analysis/02_retention_trend.py

E01b. Retention DiD: new ATS x post-AI-era interaction

term	estimate	se	ci_low	ci_high	p	n	clusters
new ATS effect, pre-AI era (2021-22)	0.003	0.009	-0.015	0.020	0.772	41677.000	40.000
DiD interaction: extra new-ATS effect, AI era	-0.023	0.010	-0.042	-0.004	0.018	41677.000	40.000
post-era gap, new vs legacy (sum of the two)	-0.021	0.009	-0.038	-0.004	0.017	41677.000	40.000

Sample: n=41677, 40 center clusters. *Notes:* Outcome: retained_6mo (0/1, linear probability). FE: center, start_month; post_ai (application-month era) entered explicitly because start-month FE do not absorb it for hires who applied before and started after the cutoff. Controls: resume/license/experience/referral/employed/wage/unemployment. Cluster: center. Row 1 = new-ATS effect before the AI era; row 2 = the DiD interaction; row 3 = their sum, the new-vs-legacy gap in the AI era. *Source:* analysis/02_retention_trend.py

E02



Sample: 41,677 permanent hires with an observed six-month outcome.
Source: analysis/02_retention_trend.py

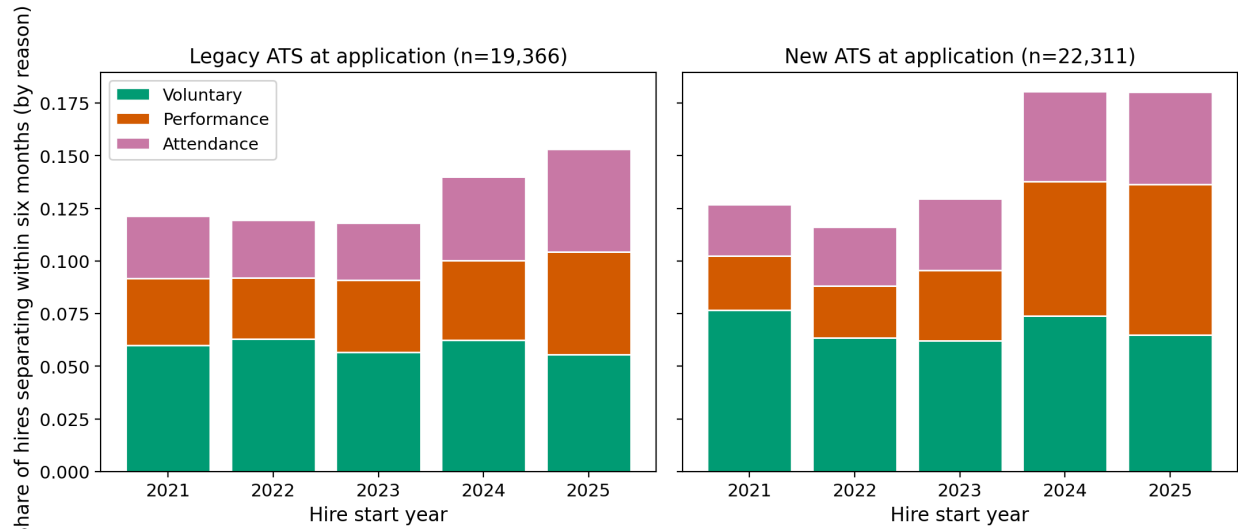
E02b. Reopen-rate DiD: new ATS x post-AI-era interaction

term	estimate	se	ci_low	ci_high	p	n	clusters
new ATS effect, pre-AI era (2021-22)	0.103	0.079	-0.051	0.257	0.189	41677.000	40.000
DiD interaction: extra new-ATS effect, AI era	0.145	0.091	-0.033	0.323	0.111	41677.000	40.000
post-era gap, new vs legacy (sum of the two)	0.248	0.090	0.071	0.425	0.006	41677.000	40.000

Sample: n=41677, 40 center clusters. *Notes:* Outcome: reopen_rate_6mo_pct (percentage points). Same FE/controls/cluster as E01b. *Source:* analysis/02_retention_trend.py

E03

E03. Separations rise after 2023 in every reason category, more on the new ATS



Sample: 41,677 permanent hires with an observed six-month outcome, by start year (2025 partial: only Jan-Jun starts have an outcome).

Source: analysis/02_retention_trend.py

E03b. Separation reason share by start year (%)

year	voluntary	performance	attendance
2021	6.320	3.080	2.850
2022	6.320	2.730	2.760
2023	5.970	3.390	3.080
2024	7.160	5.860	4.210
2025	6.360	6.890	4.430

Sample: 41,677 hires with an observed six-month outcome. Source: analysis/02_retention_trend.py

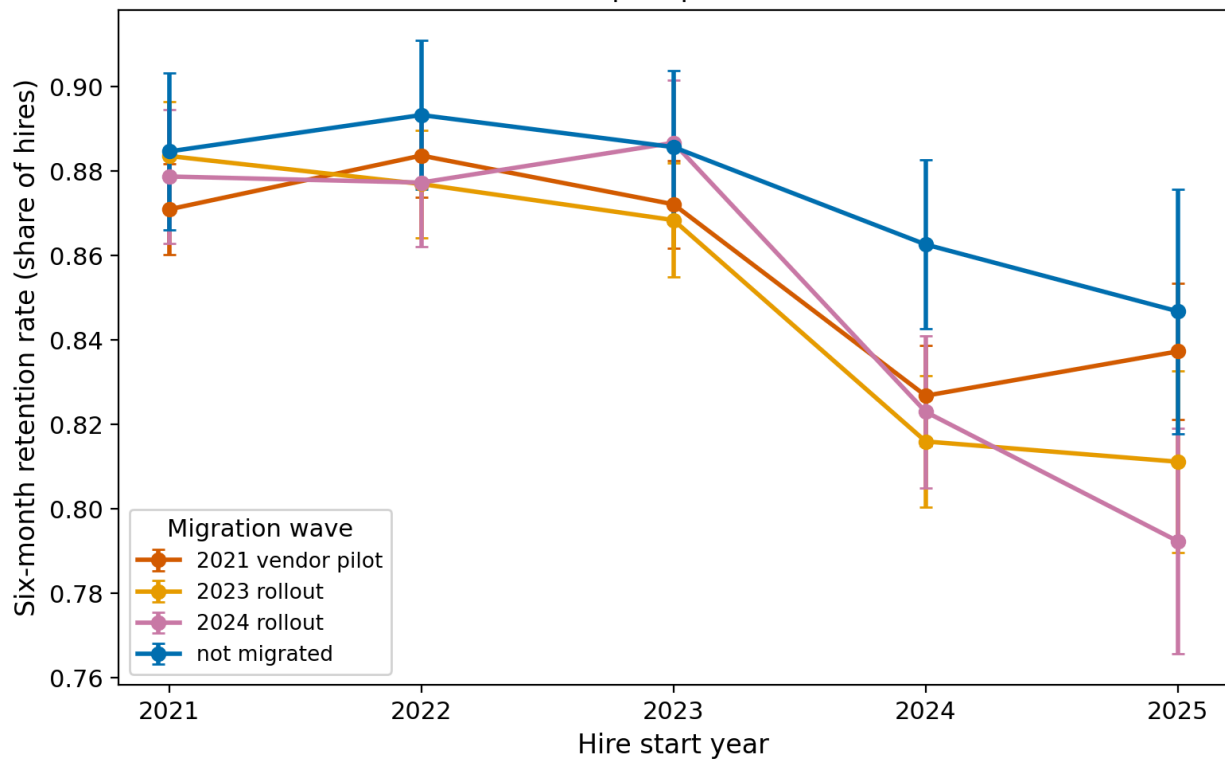
E03c. Separation-reason DiD: new ATS x post-AI-era interaction, by reason

	DiD_estimate	ci_low	ci_high
voluntary	0.010	-0.000	0.020
performance	0.010	-0.001	0.021
attendance	0.003	-0.005	0.012

Sample: n=41677, 40 center clusters. Notes: Outcome: indicator for separating within six months for the stated reason. Same spec as E01b. Source: analysis/02_retention_trend.py

E04

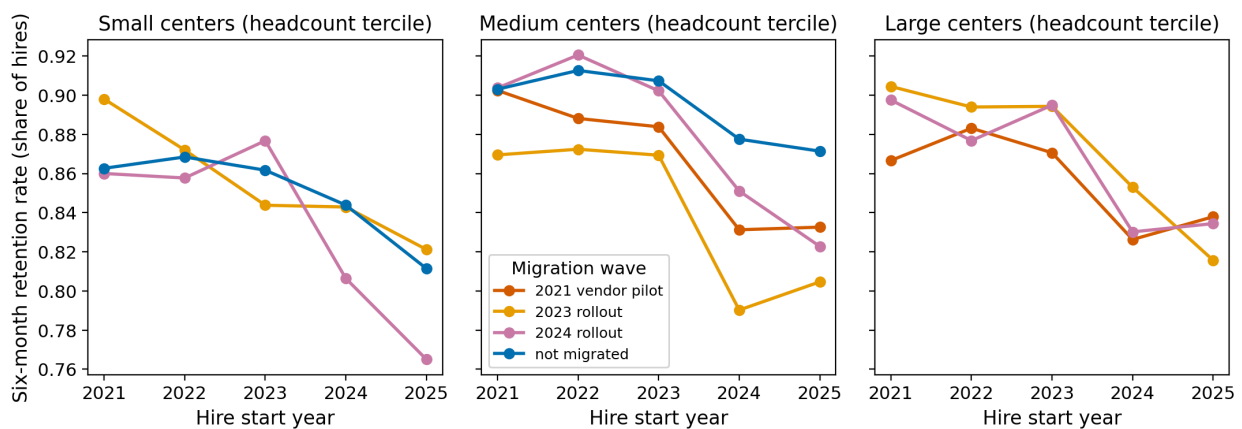
E04. The 2021 pilot centers -- the largest, best-resourced centers -- show the steepest post-2023 decline



Sample: 41,677 permanent hires with an observed six-month outcome, grouped by their center's ATS migration wave.
Source: analysis/02_retention_trend.py

E05

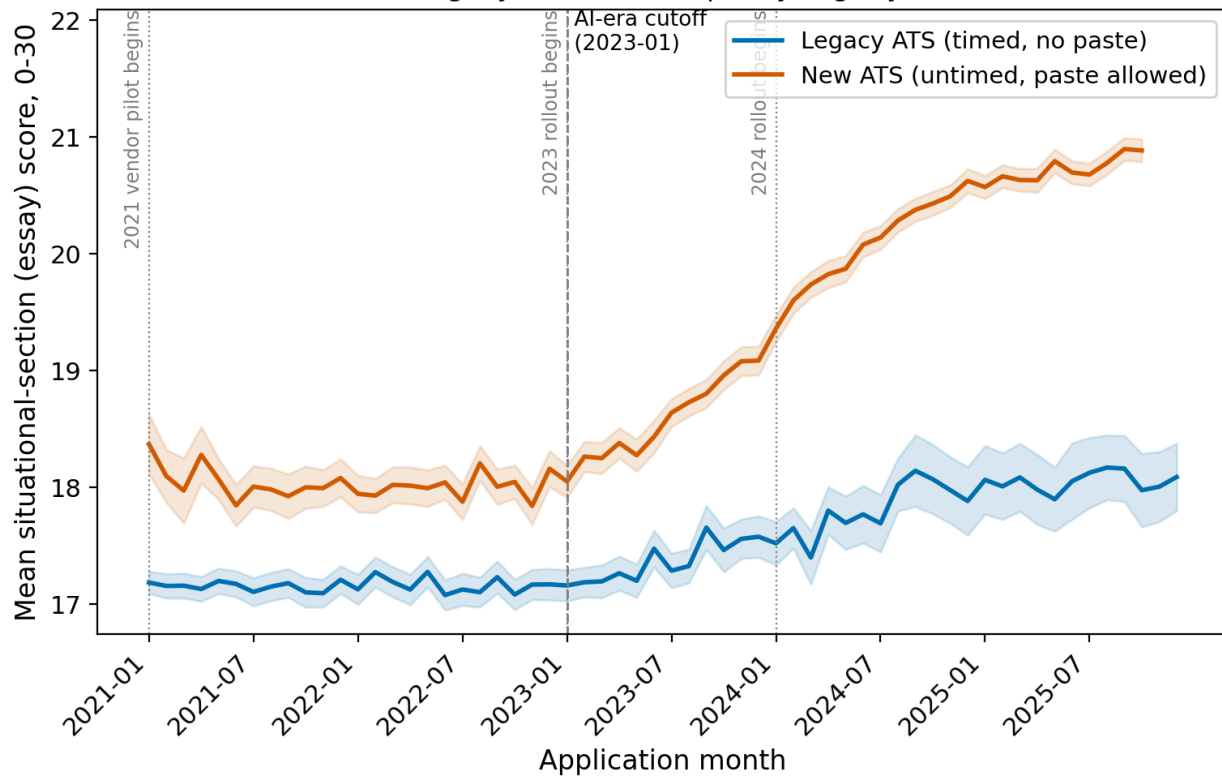
E05. The pilot-wave decline shows up within every size tercile -- size alone does not explain it



Sample: 41,677 permanent hires, split into center-headcount terciles.
Source: analysis/02_retention_trend.py

E06

E06. Essay scores climb on the new ATS from 2023;
legacy scores drift up only slightly



Sample: 529,302 reviewed applications, 2021-01 to 2025-12, 40 centers. Shaded band is a 95% CI of the monthly mean.

Source: analysis/03_pool_quality.py

E06b. Grid search for the AI-era break date in new-ATS essay scores

candidate_month	pre_mean	post_mean	t
2022-07	18.009	19.729	69.806
2022-08	18.008	19.750	73.130
2022-09	18.010	19.774	76.576
2022-10	18.001	19.799	80.367
2022-11	18.015	19.823	83.025
2022-12	18.014	19.847	86.219
2023-01	18.016	19.875	89.628
2023-02	18.008	19.900	92.929
2023-03	18.015	19.923	95.477
2023-04	18.017	19.958	99.165
2023-05	18.033	19.993	102.304
2023-06	18.045	20.028	105.414
2023-07	18.065	20.066	108.213
2023-08	18.076	20.104	111.233
2023-09	18.092	20.138	113.566
2023-10	18.125	20.181	115.857
2023-11	18.159	20.225	117.916
2023-12	18.194	20.270	119.748

Sample: Applications reviewed under the new ATS, 2021-01 to 2025-12. *Notes:* Best-fit break month by max t-stat: 2023-12. Configured cutoff in analysis/utls.py is 2023-01. *Source:* analysis/03_pool_quality.py

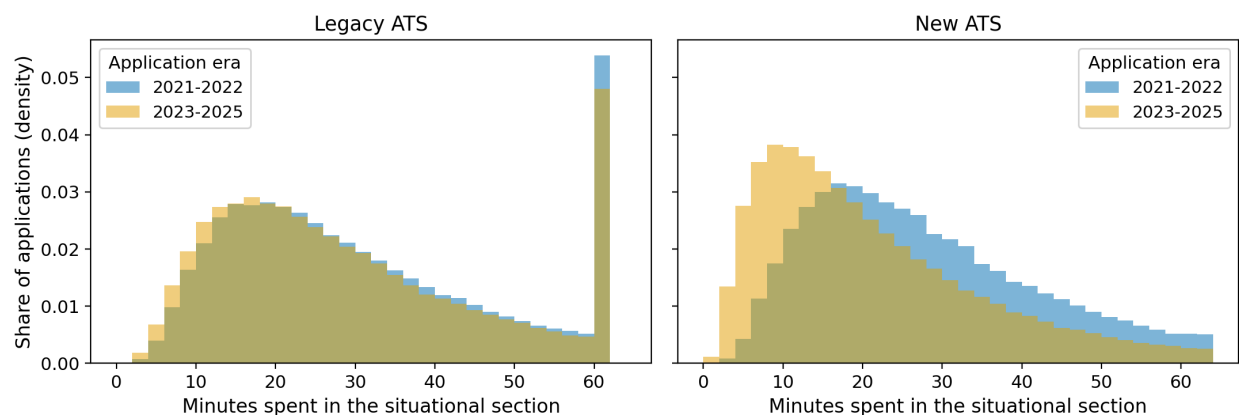
E06c. Essay-score inflation: new ATS x post-AI-era interaction

term	estimate	se	ci_low	ci_high	p	n	clusters
new ATS effect, pre-AI era (2021-22)	0.893	0.033	0.828	0.958	0.000	529302.000	40.000
DiD interaction: extra new-ATS effect, AI era	0.787	0.108	0.575	1.000	0.000	529302.000	40.000
post-era gap, new vs legacy (sum of the two)	1.680	0.114	1.457	1.903	0.000	529302.000	40.000

Sample: n=529302, 40 center clusters. *Notes:* Outcome: essay_rubric_score (0-30). Controls: resume/license/experience/referral/employed. FE: center, application_month. Cluster: center. *Source:* analysis/03_pool_quality.py

E07

E07. Time spent on the essay collapses on the new ATS after 2023; legacy shifts only slightly



Sample: 529,302 reviewed applications, split by ATS and by application era.
Source: analysis/03_pool_quality.py

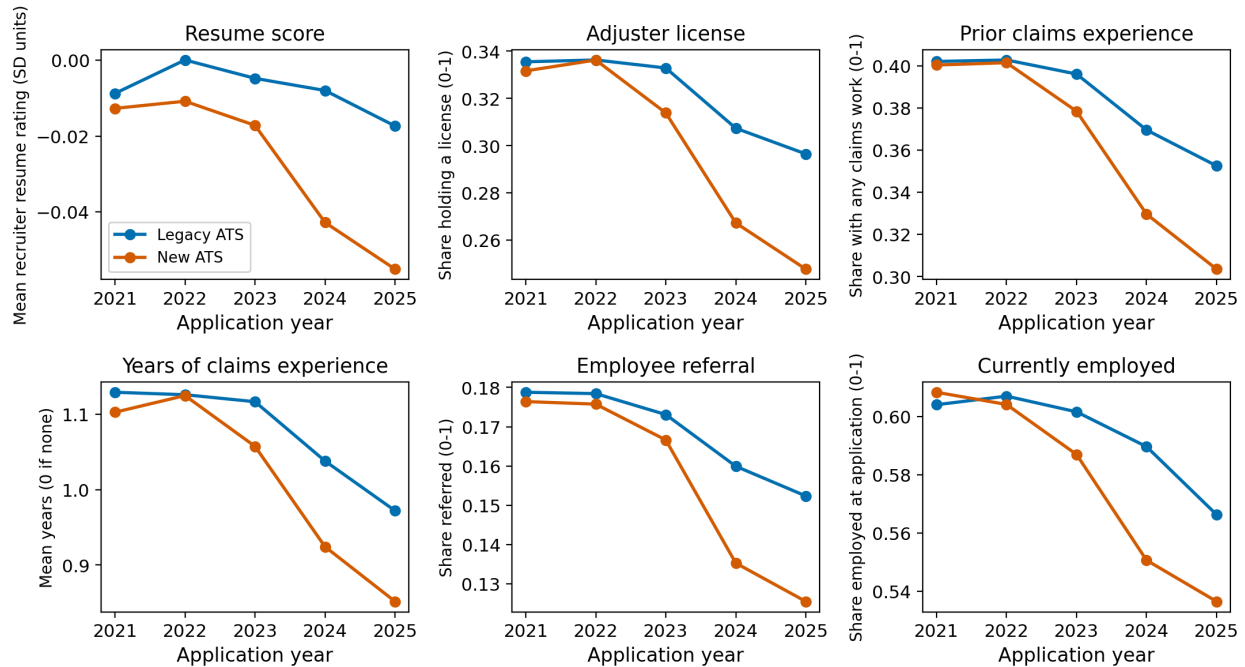
E07b. Mean essay score and share of essays under 10 minutes, by year and ATS

year	('legacy', 'essay')	('legacy', 'fast10')	('new', 'essay')	('new', 'fast10')
2021	17.152	0.061	18.025	0.059
2022	17.159	0.062	18.012	0.063
2023	17.327	0.070	18.457	0.108
2024	17.753	0.094	19.874	0.222
2025	18.050	0.117	20.694	0.283

Sample: 529,302 reviewed applications. *Notes:* The legacy (timed, no-paste) system also drifts after 2023, more slowly: a timed field slows the leak, it does not stop it. *Source:* analysis/03_pool_quality.py

E08

E08. Objective applicant attributes are flat over time --
the pool did not objectively improve while essay scores rose



Sample: 529,302 reviewed applications, yearly means by ATS at application.
Source: analysis/03_pool_quality.py

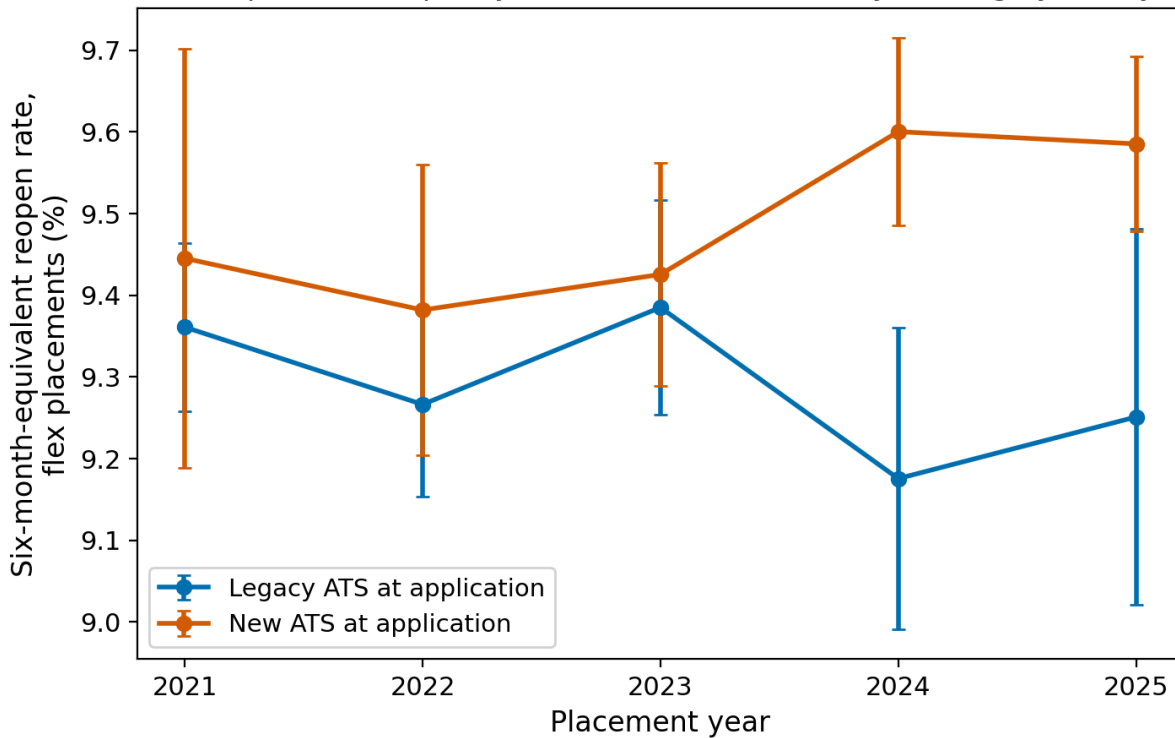
E08b. Objective applicant attributes: new ATS x post-AI-era DiD

	estimate	se	ci_low	ci_high	p	n	clusters
('resume_score', 'new ATS effect, pre-AI era (2021-22)')	-0.003	0.006	-0.015	0.008	0.559	529302.000	40.000
('resume_score', 'DiD interaction: extra new-ATS effect, AI era')	-0.010	0.006	-0.021	0.001	0.078	529302.000	40.000
('resume_score', 'post-era gap, new vs legacy (sum of the two)')	-0.013	0.007	-0.027	0.000	0.057	529302.000	40.000
('has_adjuster_license', 'new ATS effect, pre-AI era (2021-22)')	-0.000	0.003	-0.006	0.006	0.916	529302.000	40.000
('has_adjuster_license', 'DiD interaction: extra new-ATS effect, AI era')	-0.026	0.004	-0.035	-0.018	0.000	529302.000	40.000
('has_adjuster_license', 'post-era gap, new vs legacy (sum of the two)')	-0.026	0.004	-0.034	-0.019	0.000	529302.000	40.000
('prior_claims_experience', 'new ATS effect, pre-AI era (2021-22)')	0.001	0.002	-0.004	0.005	0.715	529302.000	40.000
('prior_claims_experience', 'DiD interaction: extra new-ATS effect, AI era')	-0.024	0.004	-0.032	-0.015	0.000	529302.000	40.000
('prior_claims_experience', 'post-era gap, new vs legacy (sum of the two)')	-0.023	0.004	-0.030	-0.016	0.000	529302.000	40.000
('prior_claims_years', 'new ATS effect, pre-AI era (2021-22)')	-0.006	0.009	-0.024	0.011	0.469	529302.000	40.000
('prior_claims_years', 'DiD interaction: extra new-ATS effect, AI era')	-0.053	0.013	-0.078	-0.028	0.000	529302.000	40.000
('prior_claims_years', 'post-era gap, new vs legacy (sum of the two)')	-0.060	0.011	-0.080	-0.039	0.000	529302.000	40.000
('referral', 'new ATS effect, pre-AI era (2021-22)')	-0.004	0.002	-0.007	-0.000	0.043	529302.000	40.000
('referral', 'DiD interaction: extra new-ATS effect, AI era')	-0.011	0.003	-0.017	-0.006	0.000	529302.000	40.000
('referral', 'post-era gap, new vs legacy (sum of the two)')	-0.015	0.003	-0.021	-0.009	0.000	529302.000	40.000
('currently_employed', 'new ATS effect, pre-AI era (2021-22)')	0.003	0.003	-0.003	0.009	0.342	529302.000	40.000
('currently_employed', 'DiD interaction: extra new-ATS effect, AI era')	-0.021	0.003	-0.027	-0.015	0.000	529302.000	40.000
('currently_employed', 'post-era gap, new vs legacy (sum of the two)')	-0.018	0.004	-0.025	-0.011	0.000	529302.000	40.000

Sample: n=529,302 reviewed applications, 40 center clusters. Notes: Each outcome regressed on treated_app + treated_app:post_ai with center and application-month FE, cluster center. Compare with the essay-score DiD in E06c. Source: analysis/03_pool_quality.py

E09

E09. Flex-placement quality (unselected on the essay) is roughly flat by ATS



Sample: 28,672 flex placements, 2021-01 to 2025-12. Error bars: 95% CI.
Source: analysis/03_pool_quality.py

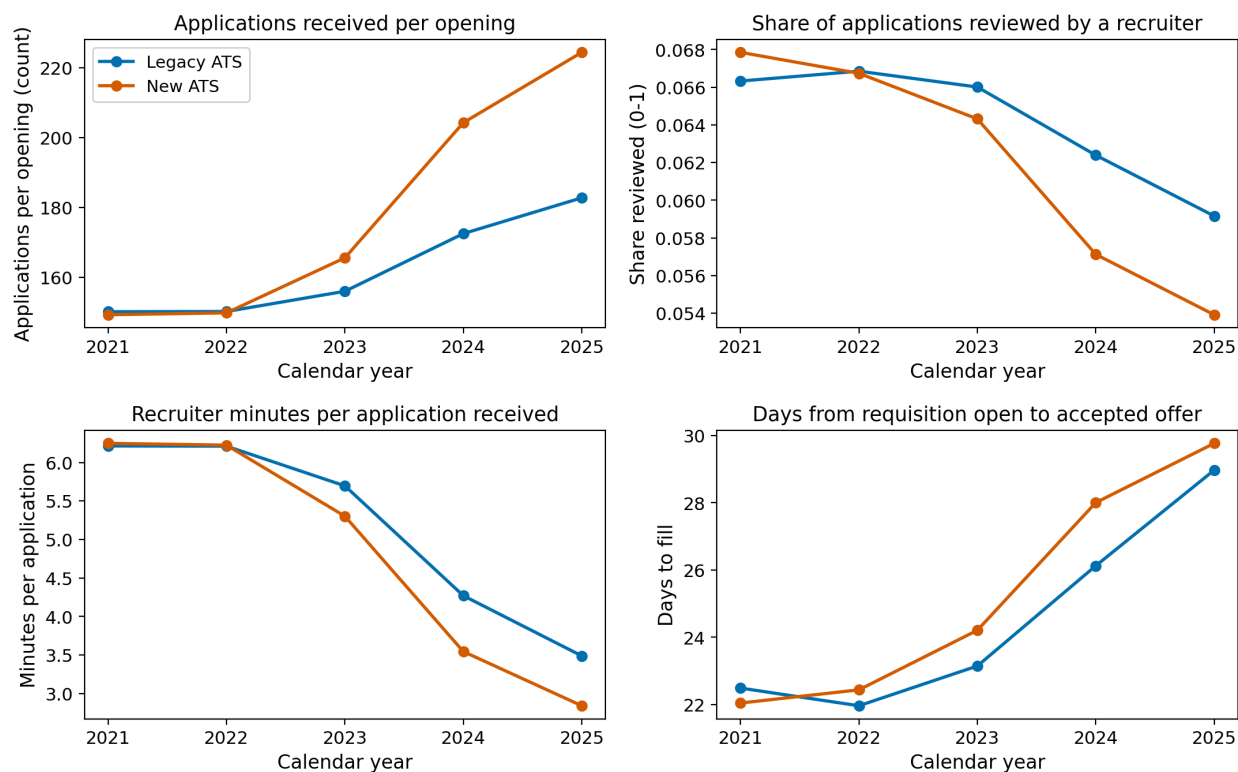
E09b. Flex reopen rate: new ATS x post-AI-era interaction

term	estimate	se	ci_low	ci_high	p	n	clusters
new ATS effect, pre-AI era (2021-22)	0.006	0.125	-0.239	0.252	0.961	28672.000	40.000
DiD interaction: extra new-ATS effect, AI era	-0.075	0.136	-0.342	0.192	0.581	28672.000	40.000
post-era gap, new vs legacy (sum of the two)	-0.069	0.085	-0.236	0.098	0.418	28672.000	40.000

Sample: n=28672, 40 center clusters. Notes: Outcome: reopen_rate_pct. Controls: certification_level FE. FE: center, placement_month. Cluster: center. Source: analysis/03_pool_quality.py

E10

E10. Recruiter capacity has not kept pace with application volume:
more applications per opening, less time per application, longer to fill



Sample: 2,400 center-months, 40 centers, 2021-01 to 2025-12, yearly means.
Source: analysis/04_recruiter_capacity.py

E10b. Recruiter-capacity metrics by year and ATS

year	ats	apps_per_opening	review_share	recruiter_minutes_per_application	days_to_fill
2021	legacy	150.200	0.070	6.210	22.490
2021	new	149.320	0.070	6.250	22.040
2022	legacy	150.260	0.070	6.210	21.960
2022	new	149.870	0.070	6.220	22.440
2023	legacy	156.040	0.070	5.700	23.150
2023	new	165.610	0.060	5.300	24.210
2024	legacy	172.530	0.060	4.270	26.130
2024	new	204.260	0.060	3.540	28.000
2025	legacy	182.770	0.060	3.490	28.970
2025	new	224.370	0.050	2.840	29.770

Sample: 2,400 center-months, yearly averages by ATS in that center-month. Source:
analysis/04_recruiter_capacity.py

E10c. Drivers of days-to-fill

	estimate	se	ci_low	ci_high	p	n	clusters
Intercept	21.374	3.894	13.742	29.006	0.000	2400.000	40.000
apps_per_opening	0.028	0.011	0.006	0.049	0.011	2400.000	40.000
recruiter_minutes_per_application	-0.792	0.384	-1.544	-0.040	0.039	2400.000	40.000
local_unemployment_rate	0.537	0.275	-0.002	1.075	0.051	2400.000	40.000

Sample: n=2400, 40 center clusters. *Notes:* Outcome: days_to_fill. FE: center, month. Cluster: center.

Source: analysis/04_recruiter_capacity.py

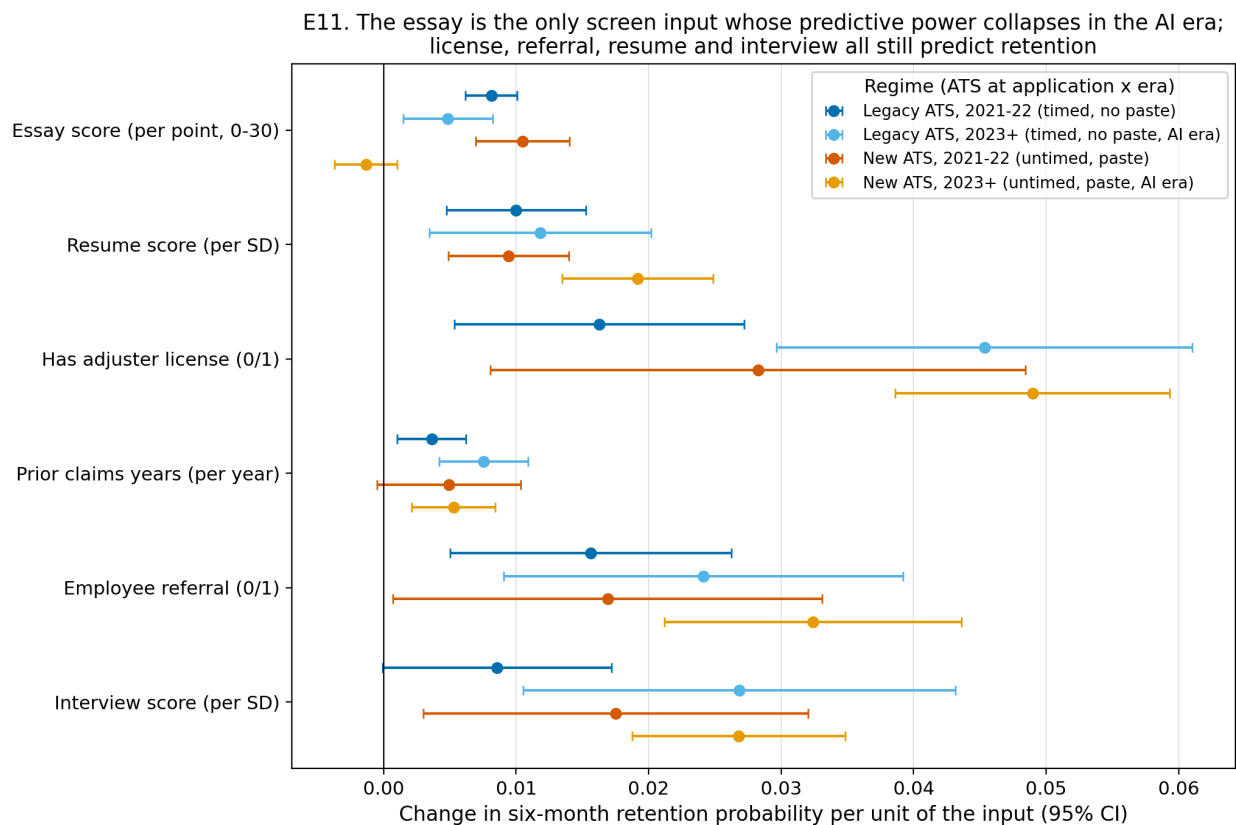
E10d. Applications received per center-month: new ATS x AI-era DiD

term	estimate	se	ci_low	ci_high	p	n	clusters
new ATS effect, pre-AI era (2021-22)	-478.320	91.000	-656.680	-299.970	0.000	2400.000	40.000
DiD interaction: extra new-ATS effect, AI era	826.120	117.500	595.820	1056.430	0.000	2400.000	40.000
post-era gap, new vs legacy (sum of the two)	347.800	126.780	99.300	596.300	0.010	2400.000	40.000

Sample: n=2400 center-months, 40 clusters. *Notes:* FE: center, month. Cluster: center. *Source:*

analysis/04_recruiter_capacity.py

E11



Sample: 41,677 hires with an observed six-month outcome, split into four cells (legacy/new ATS x application before/after 2023-01); 4,589 censored hires excluded.

Source: analysis/05_screen_components.py

E11b. Predictors of six-month retention, by regime

regime	predictor	estimate	se	ci_low	ci_high	n	clusters
legacy_pre	essay_rubric_score	0.008	0.001	0.006	0.010	13040	40
legacy_pre	resume_score	0.010	0.003	0.005	0.015	13040	40
legacy_pre	has_adjuster_license	0.016	0.006	0.005	0.027	13040	40
legacy_pre	prior_claims_years	0.004	0.001	0.001	0.006	13040	40
legacy_pre	referral	0.016	0.005	0.005	0.026	13040	40
legacy_pre	interview_score	0.009	0.004	-0.000	0.017	13040	40
legacy_post	essay_rubric_score	0.005	0.002	0.002	0.008	6326	28
legacy_post	resume_score	0.012	0.004	0.004	0.020	6326	28
legacy_post	has_adjuster_license	0.045	0.008	0.030	0.061	6326	28
legacy_post	prior_claims_years	0.008	0.002	0.004	0.011	6326	28
legacy_post	referral	0.024	0.008	0.009	0.039	6326	28
legacy_post	interview_score	0.027	0.008	0.011	0.043	6326	28
new_pre	essay_rubric_score	0.011	0.002	0.007	0.014	5920	12
new_pre	resume_score	0.009	0.002	0.005	0.014	5920	12
new_pre	has_adjuster_license	0.028	0.010	0.008	0.048	5920	12
new_pre	prior_claims_years	0.005	0.003	-0.001	0.010	5920	12
new_pre	referral	0.017	0.008	0.001	0.033	5920	12
new_pre	interview_score	0.018	0.007	0.003	0.032	5920	12
new_post	essay_rubric_score	-0.001	0.001	-0.004	0.001	16391	33
new_post	resume_score	0.019	0.003	0.013	0.025	16391	33
new_post	has_adjuster_license	0.049	0.005	0.039	0.059	16391	33
new_post	prior_claims_years	0.005	0.002	0.002	0.008	16391	33
new_post	referral	0.032	0.006	0.021	0.044	16391	33
new_post	interview_score	0.027	0.004	0.019	0.035	16391	33

Sample: 41,677 hires with an observed outcome, split into four ATS x era cells. *Notes:* Outcome: retained_6mo (linear probability). FE: center, start_month. Cluster: center. Each cell is a separate regression on its subsample; all six predictors enter together. *Source:* analysis/05_screen_components.py

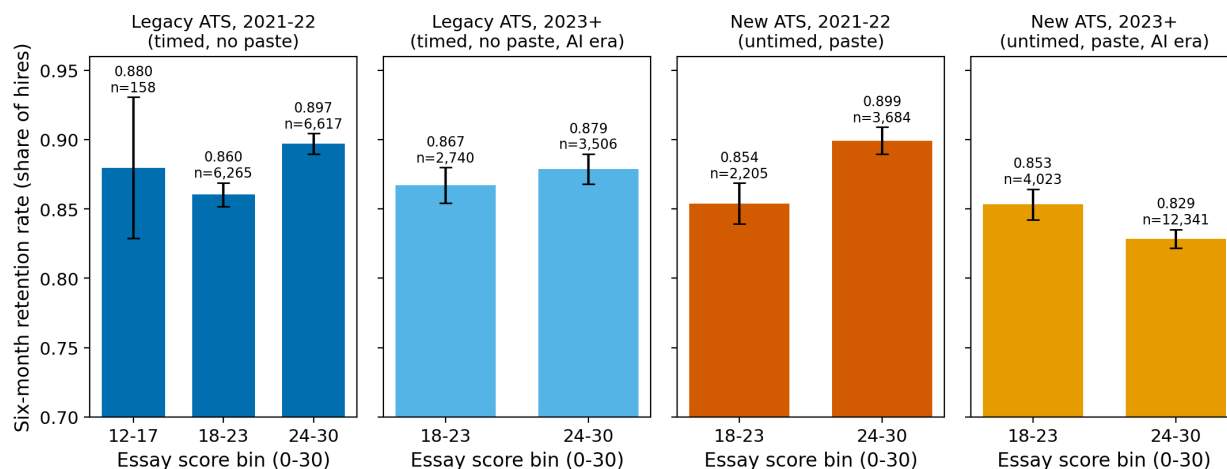
E11c. How much weight recruiters place on the essay score, by regime

regime	essay_weight_in_screen_score	se	ci_low	ci_high	n	r_squared
legacy_pre	0.250	0.000	0.250	0.251	135320	0.862
legacy_post	0.250	0.001	0.249	0.251	76215	0.848
new_pre	0.250	0.000	0.249	0.250	62232	0.861
new_post	0.249	0.000	0.249	0.250	255535	0.821

Sample: All reviewed applications, split by regime. *Notes:* Outcome: recruiter_screen_score. FE: center, application_month. Cluster: center. An R-squared near 1 means the screen score is a fixed formula of its inputs, not a judgment. *Source:* analysis/05_screen_components.py

E12

E12. A top-bin essay (24-30) used to predict better retention than an 18-23 essay;
on the new ATS after 2023 it predicts slightly worse retention



Sample: 41,677 permanent hires with an observed six-month outcome, binned by essay score within four ATS x era cells; bins with under 100 hires not shown. Error bars: 95% CI. Y-axis starts at 0.70.

Source: analysis/06_essay_validity.py

E12b. Essay-score slope on six-month retention, by regime (hires)

regime	estimate	se	ci_low	ci_high	n_cell
legacy_pre	0.009	0.001	0.007	0.010	13040
legacy_post	0.005	0.002	0.001	0.008	6326
new_pre	0.011	0.002	0.007	0.014	5920
new_post	-0.003	0.001	-0.005	-0.000	16391

Sample: n=41,677, 40 center clusters. Notes: Outcome: retained_6mo. Controls: resume, license, prior years, referral, interview score. FE: center, start_month. Cluster: center. Hires are selected on the essay, which attenuates every slope toward zero. Source: analysis/06_essay_validity.py

E12c. Essay-score slope on six-month reopen rate, by regime (hires)

regime	estimate	se	ci_low	ci_high	n_cell
legacy_pre	-0.147	0.010	-0.167	-0.127	13040
legacy_post	-0.093	0.015	-0.123	-0.063	6326
new_pre	-0.180	0.016	-0.212	-0.148	5920
new_post	0.035	0.010	0.016	0.054	16391

Sample: n=41,677, 40 center clusters. Notes: Outcome: reopen_rate_6mo_pct (percentage points).

Negative = higher essay score predicts fewer reopens. Same FE/cluster as E12b. Source:

analysis/06_essay_validity.py

E12d. Essay-score slope on reopen rate, by regime (flex placements -- NOT selected on the essay by Meridian)

regime	estimate	se	ci_low	ci_high	n_cell	weighted_estimate	weighted_se
legacy_pre	-0.179	0.010	-0.199	-0.160	9079	-0.177	0.010
legacy_post	-0.122	0.012	-0.146	-0.098	5541	-0.118	0.013
new_pre	-0.207	0.019	-0.244	-0.169	2346	-0.212	0.021
new_post	-0.069	0.007	-0.083	-0.056	11706	-0.069	0.007

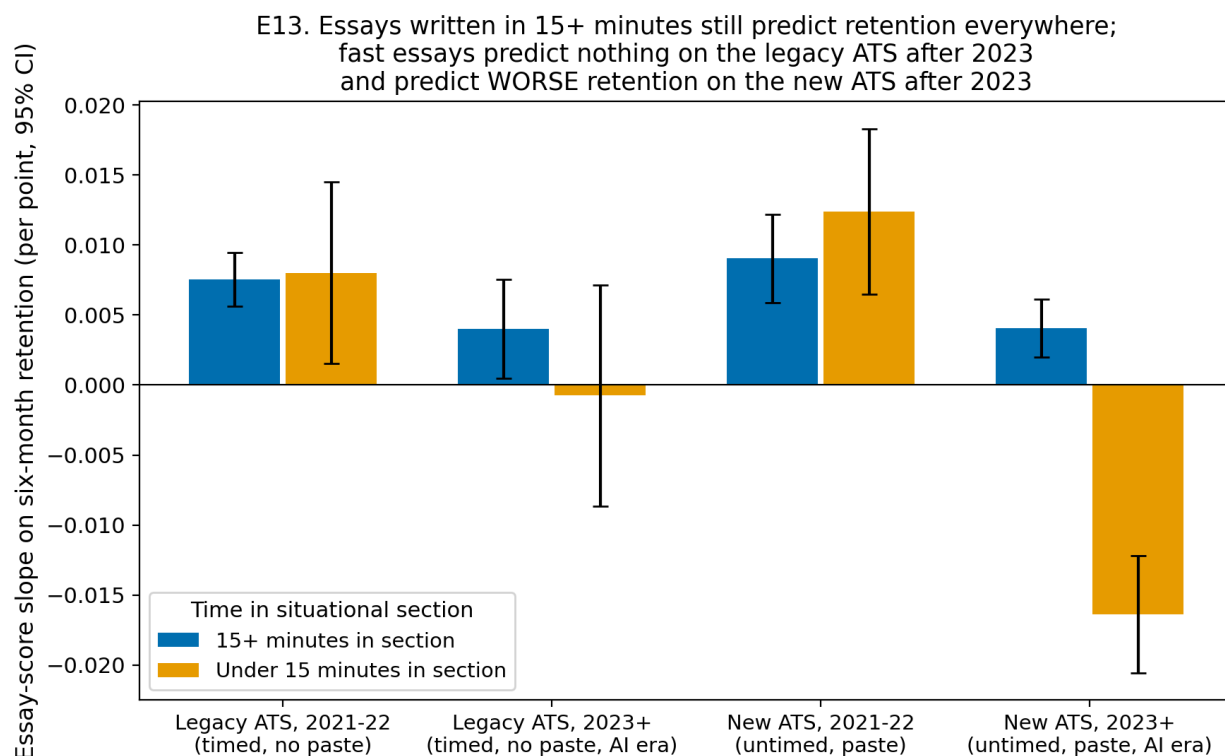
Sample: n=28,672 flex placements, 40 center clusters. *Notes:* Outcome: reopen_rate_pct. Controls: resume, license, prior years, referral, certification level. FE: center, placement_month. Cluster: center. A MORE NEGATIVE slope means the essay is more informative. Weighted columns weight by claims_closed. *Source:* analysis/06_essay_validity.py

E12e. Spec C: essay slope on retention in the 12 months before vs after a center's own ATS migration (2023/24 waves) vs never-migrated centers over the same calendar window

group	window	estimate	se	ci_low	ci_high	n
2023/24-wave centers, own migration month	12 months before	0.006	0.002	0.002	0.010	8327
2023/24-wave centers, own migration month	12 months after	-0.002	0.002	-0.006	0.003	8327
never-migrated centers, pseudo-event 2023-09	12 months before	0.007	0.003	0.002	0.012	2359
never-migrated centers, pseudo-event 2023-09	12 months after	-0.003	0.002	-0.007	0.001	2359

Sample: 8,327 hires at 2023/2024-wave centers; 2,359 hires at never-migrated centers. *Notes:* Same controls/FE/cluster as E12b. The never-migrated slope also falls over this window (calendar 2022-09 to 2024-08), so the migration itself is not the whole story: the AI era degrades the timed section too, more slowly. *Source:* analysis/06_essay_validity.py

E13



Sample: 41,677 hires with an observed six-month outcome, split by ATS x era cell and by time spent in the situational section.

Source: analysis/06_essay_validity.py

E13b. Essay-score slope on retention, by regime and by whether the applicant spent under 15 minutes in the situational section

regime	fast_lt_15min	estimate	se	ci_low	ci_high	n_cell
legacy_pre	0	0.008	0.001	0.006	0.009	11699
legacy_pre	1	0.008	0.003	0.002	0.014	1341
legacy_post	0	0.004	0.002	0.000	0.008	5270
legacy_post	1	-0.001	0.004	-0.009	0.007	1056
new_pre	0	0.009	0.002	0.006	0.012	5353
new_pre	1	0.012	0.003	0.006	0.018	567
new_post	0	0.004	0.001	0.002	0.006	9876
new_post	1	-0.016	0.002	-0.021	-0.012	6515

Sample: 41,677 hires with an observed outcome. Notes: Outcome: retained_6mo. essay x cell x fast15 slopes with cell x fast15 main effects, center + start-month FE, cluster center. fast_lt_15min=1 means essay_section_minutes < 15. Source: analysis/06_essay_validity.py

E13c. Essay score per minute spent in the section, by regime

regime	estimate	se	n
legacy_pre	0.054	0.001	135320.000
legacy_post	0.030	0.003	76215.000
new_pre	0.034	0.001	62232.000
new_post	-0.021	0.001	255535.000

Sample: All reviewed applications, by cell. *Notes:* essay_rubric_score ~ essay_section_minutes + center FE + application-month FE, cluster center. Positive = longer effort earns a higher score. *Source:* analysis/06_essay_validity.py

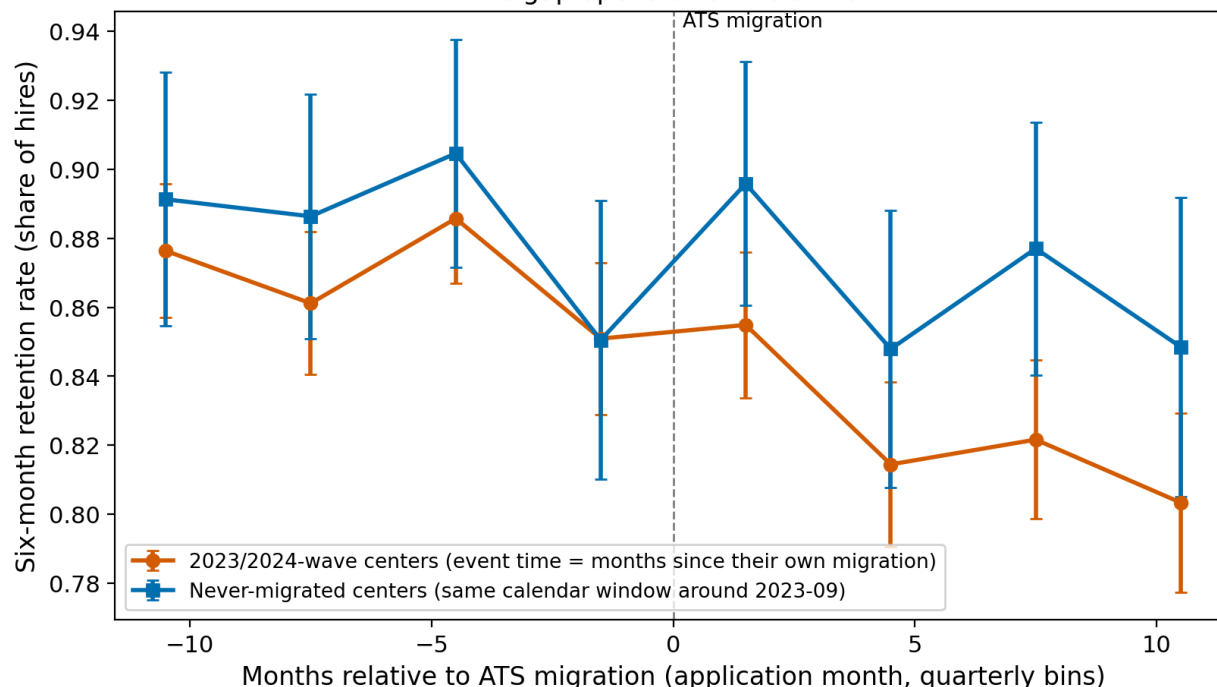
E14. Placebo test: slope of other screen inputs on six-month retention, by regime

regime	estimate	se	ci_low	ci_high	variable
legacy_pre	0.009	0.003	0.004	0.014	resume_score
legacy_post	0.012	0.004	0.003	0.020	resume_score
new_pre	0.008	0.002	0.003	0.013	resume_score
new_post	0.023	0.003	0.017	0.028	resume_score
legacy_pre	0.020	0.006	0.009	0.030	has_adjuster_license
legacy_post	0.050	0.009	0.033	0.066	has_adjuster_license
new_pre	0.033	0.010	0.014	0.052	has_adjuster_license
new_post	0.060	0.005	0.050	0.071	has_adjuster_license
legacy_pre	0.004	0.001	0.001	0.007	prior_claims_years
legacy_post	0.009	0.002	0.005	0.012	prior_claims_years
new_pre	0.005	0.003	0.000	0.011	prior_claims_years
new_post	0.008	0.002	0.005	0.011	prior_claims_years
legacy_pre	-0.005	0.004	-0.013	0.004	interview_score
legacy_post	0.019	0.008	0.003	0.035	interview_score
new_pre	0.001	0.007	-0.013	0.014	interview_score
new_post	0.032	0.004	0.024	0.040	interview_score

Sample: 41,677 hires with an observed outcome (interview_score restricted to applicants who were interviewed). *Notes:* Univariate slopes (each input alone) with center + start-month FE, cluster center. If these are stable while the essay's slope collapses (E12b), the essay result is not an artifact of a center-wide shift in hiring quality or measurement. E11b gives the multivariate version. *Source:* analysis/06_essay_validity.py

E15

E15. Migrating and never-migrated centers track each other before migration; the gap opens after the switch



Sample: 8,327 hires at 2023/2024-migration-wave centers (-12 to +11 months around each center's migration) and 2,359 hires at never-migrated centers (same window around the waves' median migration month). Error bars: 95% CI.
Source: analysis/07_identification.py

E15b. Event-study means: retention before/after migration vs never-migrated

period	migrating_centers	never_migrated	gap_never_minus_migrating
12 months before	0.869	0.883	0.014
12 months after	0.825	0.868	0.043

Sample: 8,327 + 2,359 hires, event window +/-12 months. Source: analysis/07_identification.py

E15c. Selection on the essay into the hire sample, by regime

regime4	applicant_essay_mean	applicant_essay_sd	hire_essay_mean	hire_essay_sd	hire_rate_bin_18_23	hire_rate_bin_24_30	share_applicants_24_30
legacy_pre	17.155	4.607	23.653	2.957	0.120	0.579	0.084
legacy_post	17.558	4.740	24.005	2.975	0.097	0.486	0.105
new_pre	18.016	4.591	24.502	2.869	0.083	0.501	0.118
new_post	19.875	4.832	25.552	2.845	0.041	0.274	0.232

Sample: All reviewed applications and all hires with an observed outcome. Notes: If the hire-sample essay SD were much smaller in one cell, a zero slope there could be a range-restriction artifact. It is not: the SD is similar in every cell. What changed is the share of applicants scoring 24-30 and the hire rate within that bin. Source: analysis/07_identification.py

E15d. Small-cluster inference: wild-cluster bootstrap on the headline essay-slope differences

test	difference	cluster_ se	cluster_ p	wild_ bootstrap_ p_299	n	clusters
hires: retention slope, new_post minus legacy_pre	-0.011	0.001	0.000	0.000	41677.000	40.000
hires: retention slope, legacy_post minus legacy_pre	-0.004	0.002	0.063	0.080	41677.000	40.000
flex: reopen slope, new_post minus legacy_pre	0.110	0.013	0.000	0.000	28672.000	40.000
flex: reopen slope, legacy_post minus legacy_pre	0.057	0.016	0.000	0.003	28672.000	40.000

Sample: Hires with an observed outcome (retention) and flex placements (reopen). *Notes:* H0: the essay slope in the named cell equals the legacy pre-2023 slope. Rademacher wild cluster bootstrap, 299 draws, 40 center clusters, restricted model imposes H0. A p below 0.05 means the difference survives small-cluster inference. *Source:* analysis/07_identification.py

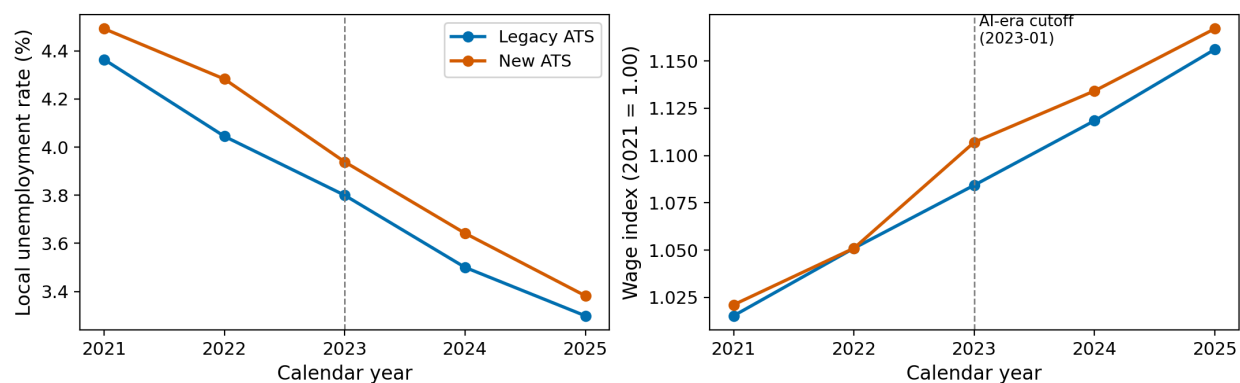
E15e. Share of the legacy pre-2023 essay signal retained, by cell: hires sample vs flex sample

regime	hires_share_of_legacy_pre_slope_ retained	flex_share_of_legacy_pre_slope_ retained	range_retained
legacy_pre	1.000	1.000	[100%, 100%]
legacy_post	0.547	0.681	[55%, 68%]
new_pre	1.232	1.153	[115%, 123%]
new_post	0.000	0.387	[0%, 39%]

Sample: Slopes from E12b (hires) and E12d (flex). Negative hire slopes are clipped at 0% retained. *Notes:* The two samples bracket the truth: hires are selected on the essay (attenuated, lower bound on retained signal); flex are not. *Source:* analysis/07_identification.py

E16

E16. Local labor-market conditions move together regardless of ATS;
they do not explain the new-vs-legacy divergence



Sample: 2,400 center-months, 40 centers, yearly means by ATS in that center-month; dashed line = 2023 AI-era cutoff. *Source:* analysis/07_identification.py

E16b. Labor-market conditions by year and ATS

year	ats	local_unemployment_rate	wage_index
2021	legacy	4.364	1.015
2021	new	4.492	1.021
2022	legacy	4.045	1.051
2022	new	4.283	1.051
2023	legacy	3.800	1.084
2023	new	3.939	1.107
2024	legacy	3.500	1.118
2024	new	3.642	1.134
2025	legacy	3.299	1.156
2025	new	3.382	1.167

Sample: 2,400 center-months. *Source:* analysis/07_identification.py

E17. Cost per retained employee: status quo vs counterfactuals where the essay still screened

scenario	n	r	days_to_fill	reopen_pct	C_hire	C_sep	C_policy	C_quality	CPR_excl_quality	CPR_incl_quality
Status quo: new ATS, hires starting 2024-25 (observed)	11440.000	0.820	28.930	8.918	18699.000	2540.000	0.000	17531.000	23377.000	40908.000
Status quo (alt): new ATS, all AI-era hires 2023-25 (observed)	16391.000	0.834	28.930	8.592	18699.000	2540.000	0.000	16888.000	22916.000	39804.000
Counterfactual A: legacy centers, hires starting 2024-25 (observed)	2487.000	0.857	27.260	7.985	18334.000	2540.000	0.000	15695.000	21821.000	37516.000
Counterfactual A': status quo + DiD retention gain, status-quo days to fill	—	0.843	28.930	7.985	18699.000	2540.000	0.000	15695.000	22665.000	38360.000
Counterfactual B: new ATS before the AI era (2021-22), costed at 2025 rates	5920.000	0.882	28.930	7.645	18699.000	2540.000	0.000	15027.000	21547.000	36574.000

Sample: Hires with an observed six-month outcome, by scenario; all costed at 2025 average rates (E17a). CPR = cost per retained employee, \$. *Notes:* A = the seven never-migrated centers (different, smaller centers). A' = the status-quo cohort with retention raised by the H1 DiD gain (E01b) and reopen set to the legacy level. B = the same new-ATS centers before AI writing. C_hire excludes recruiter screening time (E17b). *Source:* analysis/08_cost_worksheet.py

E17a. Worksheet inputs: 2025 average rates across 40 centers vs the prompt's worked example

	2025 average	prompt_worked_example
sourcing_cost_per_opening	746.910	747.000
training_cost_per_hire	5085.230	5085.000
ramp_productivity_loss_per_hire	6540.390	6540.000
vacancy_cost_per_day	218.650	219.000
separation_cost_per_exit	2539.850	2540.000
claims_per_adjuster_6mo	608.550	609.000
cost_per_reopened_claim	323.010	323.000
recruiter_hourly_rate	66.530	—

Sample: 480 center-months in 2025. *Source:* analysis/08_cost_worksheet.py

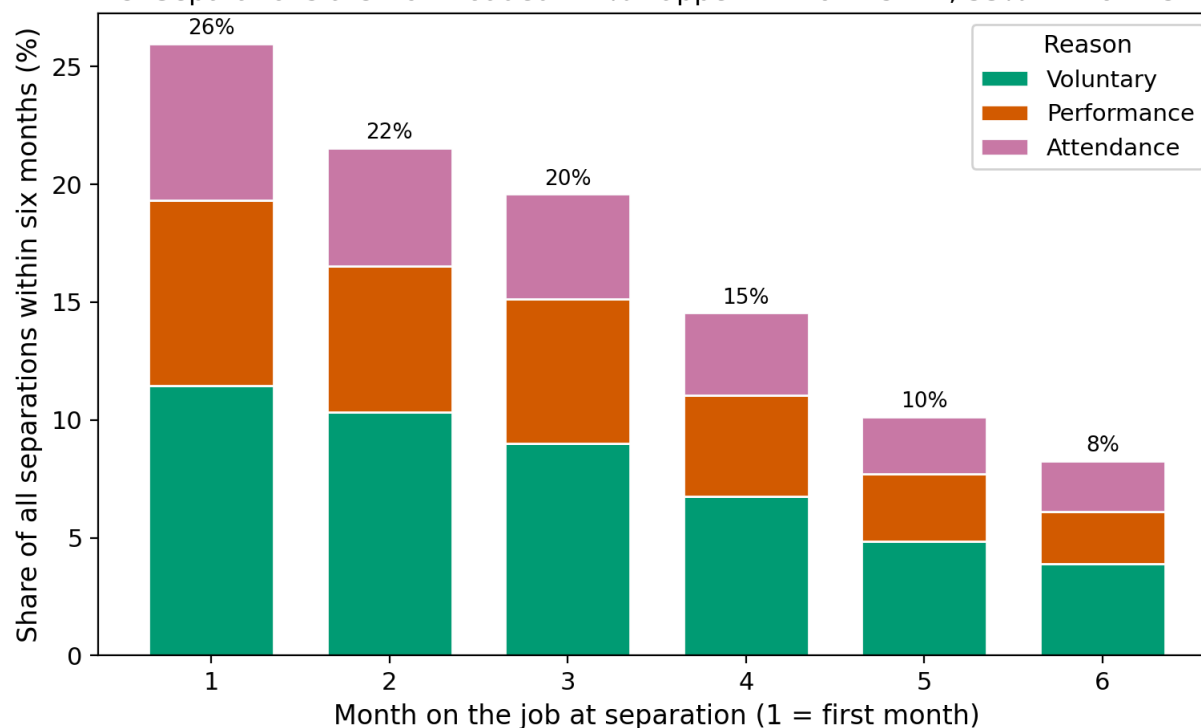
E17b. Recruiter screening time per hire (not in the prompt's C_hire; shown for option c)

	value
recruiter_minutes_per_application_2025	2.954
applications_received_per_hire_2025 (all centers, annualised)	226.045
recruiter_hourly_rate_2025	66.529
recruiter_screening_cost_per_hire_\$	740.359
as_share_of_C_hire_status_quo	0.040

Sample: 2025 center-months and 2025 hires (H1 x 2). *Source:* analysis/08_cost_worksheet.py

E18

E18. Separations are front-loaded: 47% happen in months 1-2, 33% in months 4-6



Sample: 5,796 hires who separated within six months, out of 41,677 with an observed outcome (all regimes).

Source: analysis/08_cost_worksheet.py

E18b. Separations by month on the job and reason (counts)

months_to_sep	attendance	performance	voluntary	total
1	385	457	663	1505
2	291	359	598	1248
3	258	355	522	1135
4	203	249	391	843
5	140	166	281	587
6	124	129	225	478

Sample: 5,796 hires who separated within six months. Source: analysis/08_cost_worksheet.py

E19. Cost per retained employee by option, vs status quo

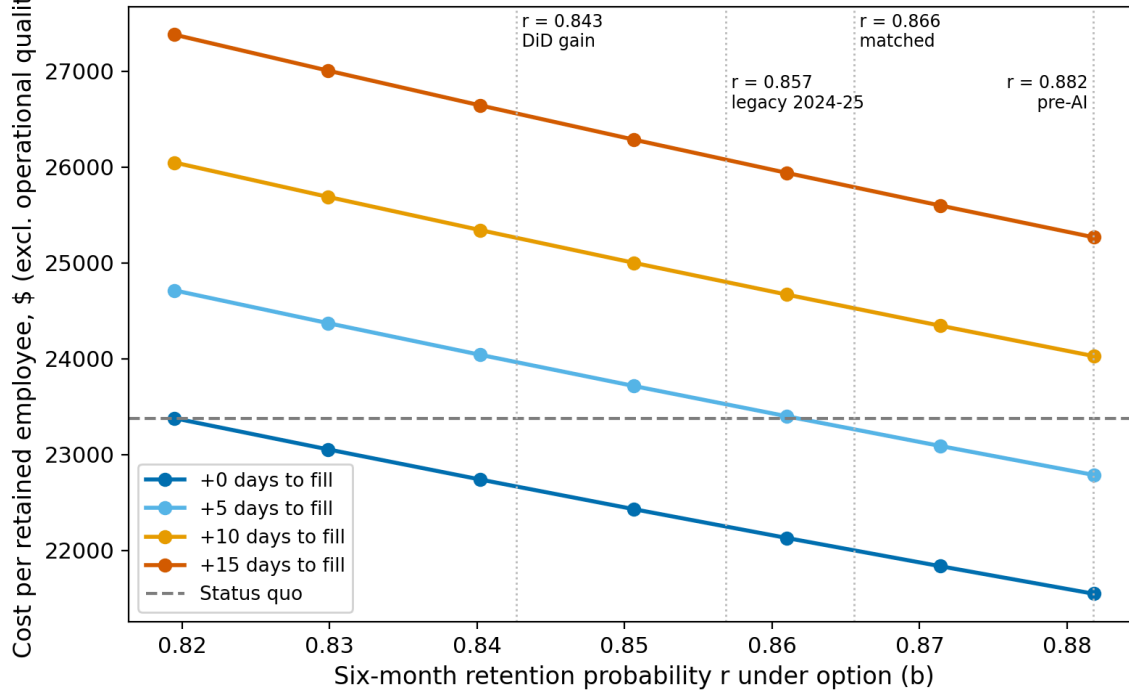
scenario	n	r	days_to_fill	reopen_pct	C_hire	C_sep	C_policy	C_quality	CPR_excl_quality	CPR_incl_quality	delta_vs_status_quo_excl_quality	annual_delta_excl_quality_\$M	delta_vs_status_quo_incl_quality	annual_delta_incl_quality_\$M
Status quo: new ATS, hires starting 2024-25 (observed)	11440.000	0.820	28.930	8.918	18699.000	2540.000	0.000	17531.000	23377.000	40908.000	0.000	0.000	0.000	0.000
(a) Do nothing: status quo continues	11440.000	0.820	28.930	8.918	18699.000	2540.000	0.000	17531.000	23377.000	40908.000	0.000	0.000	0.000	0.000
(a') Do nothing, 2024->25 retention change (+0.000) continues one more year	—	0.820	28.930	8.918	18699.000	2540.000	0.000	17531.000	23354.000	40885.000	-22.879	-0.160	-22.879	-0.160
(b, low) Applicants bear it: retention gain = H1 DiD only; reopen = legacy	—	0.843	28.930	7.985	18699.000	2540.000	0.000	15695.000	22665.000	38360.000	-712.562	-4.870	-2548.214	-17.400
(b) Applicants bear it: timed/ no-paste section; r and reopen = legacy centers 2024-25; status-quo days to fill	2487.000	0.857	28.930	7.985	18699.000	2540.000	0.000	15695.000	22247.000	37942.000	-1130.102	-7.720	-2965.754	-20.260
(b, high) Applicants bear it: legacy r, reopen AND legacy days to fill	2487.000	0.857	27.260	7.985	18334.000	2540.000	0.000	15695.000	21821.000	37516.000	-1556.492	-10.630	-3392.144	-23.170
(b, matched) Applicants bear it: r and reopen = status quo + legacy gap matched on size and region (nearest neighbour, k=2, on size_index within region, E15f); status-quo days to fill	—	0.866	28.930	8.035	18699.000	2540.000	0.000	15794.000	21998.000	37792.000	-1378.993	-9.420	-3115.888	-21.280
(c) Meridian bears it: structured interview; ASSUMED r recovers 50% of the legacy gap,	—	0.838	31.930	8.452	19559.000	2540.000	0.000	16613.000	23826.000	40439.000	448.467	3.060	-469.359	-3.210

scenario	n	r	days_to_fill	reopen_pct	C_hire	C_sep	C_policy	C_quality	CPR_excl_quality	CPR_incl_quality	delta_vs_status_quo_excl_quality	annual_delta_excl_quality_\$M	delta_vs_status_quo_incl_quality	annual_delta_incl_quality_\$M
+15 recruiter min per reviewed application (\$204/hire), +3 days to fill (d) Bear it after hiring: \$500 retention bonus at six months; ASSUMED r recovers 17% of the legacy gap (separations are front-loaded, E18)	—	0.826	28.930	8.918	18699.000	2540.000	500.000	17531.000	23682.000	41213.000	304.548	2.080	304.548	2.080

Sample: Status quo, (a) and (b) are observed cohorts; (a'), (b low/high/matched), (c) and (d) rest on the stated assumptions in the row label. Annual figures use 6,830 retained seats a year (8,334 new-ATS hires x r, 2025 H1 pace x 2). *Notes:* Negative delta = cheaper than status quo. (c) and (d) are not estimated from Meridian data because Meridian has not run them; their retention effects are assumptions and E20 shows how far they can move before the ranking changes. *Source:* analysis/08_cost_worksheet.py

E20

E20. Option (b) stays cheaper than the status quo unless a timed section adds roughly a week of vacancy per hire without buying the full legacy retention gain



Sample: Worksheet evaluated over a grid of r (status quo to pre-AI level) and extra days to fill; other inputs at 2025 averages; quality excluded.

Source: analysis/08_cost_worksheet.py

E20b. Sensitivity table: cost per retained employee under option (b), \$ excl. quality, by r (rows) and extra days to fill (columns)

r	0	5	10	15
0.820	23377.000	24711.000	26045.000	27379.000
0.830	23052.000	24370.000	25687.000	27004.000
0.840	22739.000	24040.000	25341.000	26642.000
0.851	22430.000	23715.000	25000.000	26285.000
0.861	22128.000	23398.000	24667.000	25937.000
0.871	21834.000	23088.000	24343.000	25597.000
0.882	21546.000	22786.000	24026.000	25265.000

Sample: 2025-average cost rates. Source: analysis/08_cost_worksheet.py

E20c. Break-even: extra vacancy days that erase option (b)'s saving; recovery share option (c) needs

r_ assumed	label	breakeven_ extra_ days_excl_ quality	breakeven_ extra_ days_incl_ quality	option_c_ recovery_ share_ needed_ to_break_ even_ excl_ quality	at_extra_ recruiter_ cost_per_ hire_\$	at_ extra_ days_ to_fill
0.843	DiD gain only	2.746	9.820	0.890	204.101	3.000
0.857	legacy 2024-25 observed	4.429	11.622	—	—	—
0.866	legacy gap matched on size and region (nn2, E15f)	5.459	12.725	—	—	—
0.882	pre-AI level	7.380	14.783	—	—	—

Sample: 2025-average rates; status quo from E19 row 0. *Notes:* Each extra day to fill costs vacancy_cost_per_day / r per retained employee. *Source:* analysis/08_cost_worksheet.py